INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR

DHV PROJECT REPORT

Metro train Ridership Analysis

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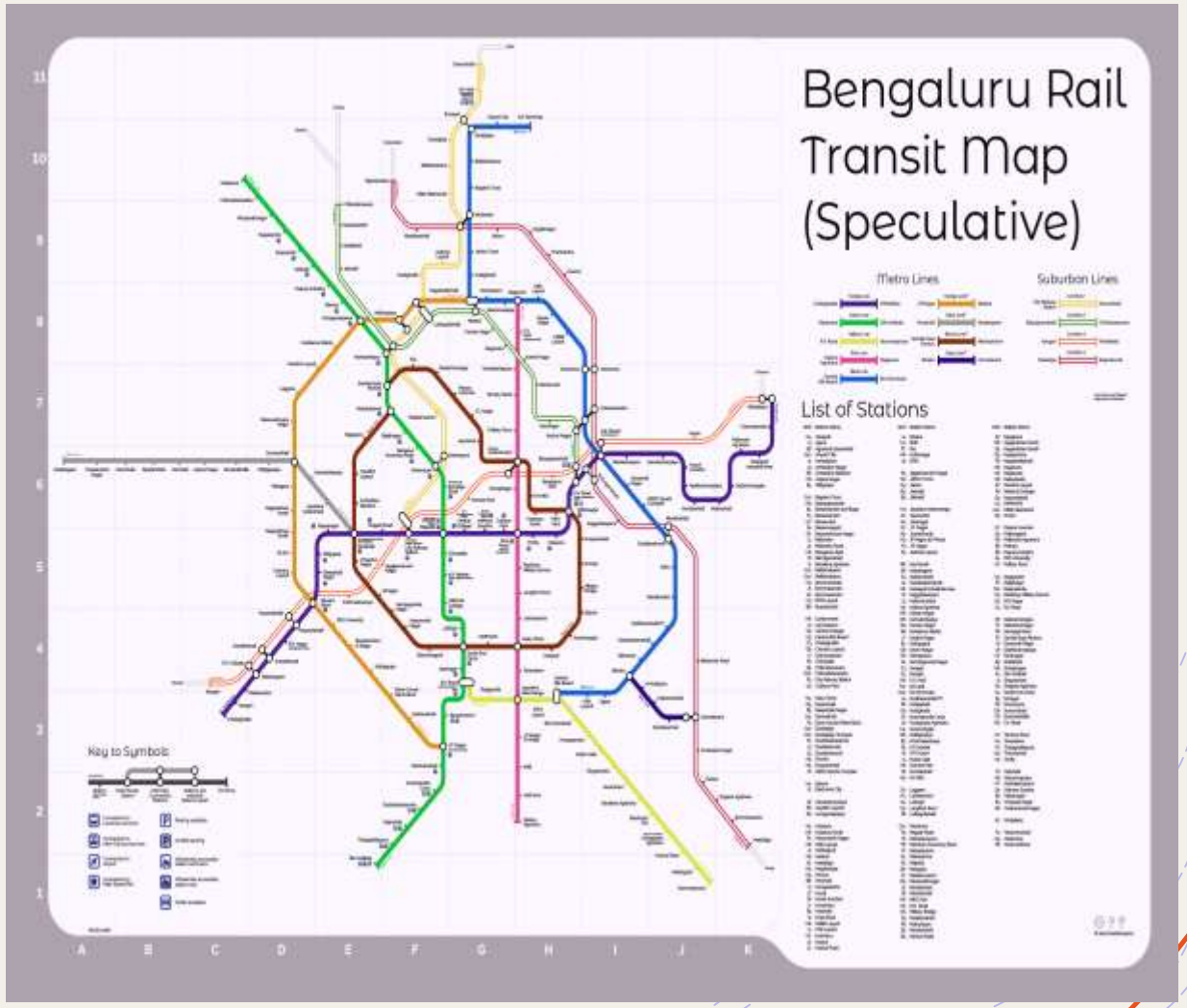
OVERVIEW

The Bangalore Metro Rail Corporation Limited (BMRCL) publishes daily ridership data every 24 hours. Unfortunately, they do not provide historical data beyond one day. This Project contains data to automate the process of maintaining official ridership data from BMRCL by web scrapping. The growing dataset and its analysis opens a window of insight into the throbbing pulse of Bangalore.

The primary goal is to analyze Metro Ridership data to identify patterns and visualize them using software like PowerBI,Seaborn



ABOUT DATASET

NammaMetro_Ridership_Dataset.csv is updated daily with each row representing the previous day's ridership stats. BMRCL's Ridership page offers the following data points:

- Record Date
- Total Smart Cards (= Stored Value Card + One Day Pass + Three Day Pass + Five Day Pass)
- Tokens, Total NCMC, Group Ticket
- Total QR (= QR NammaMetro + QR WhatsApp + QR Paytm)

The first entry of this dataset was recorded on 2024-10-26. As the dataset grows, one day and one row at a time, it will become a valuable resource for anyone interested in transportation trends and urban studies.

One Day On NammaMetro

Missing Days

Daily Overall Trend

The Dataset At A Glance

Missing Days

1.How are missing records identified and handled before visualization?

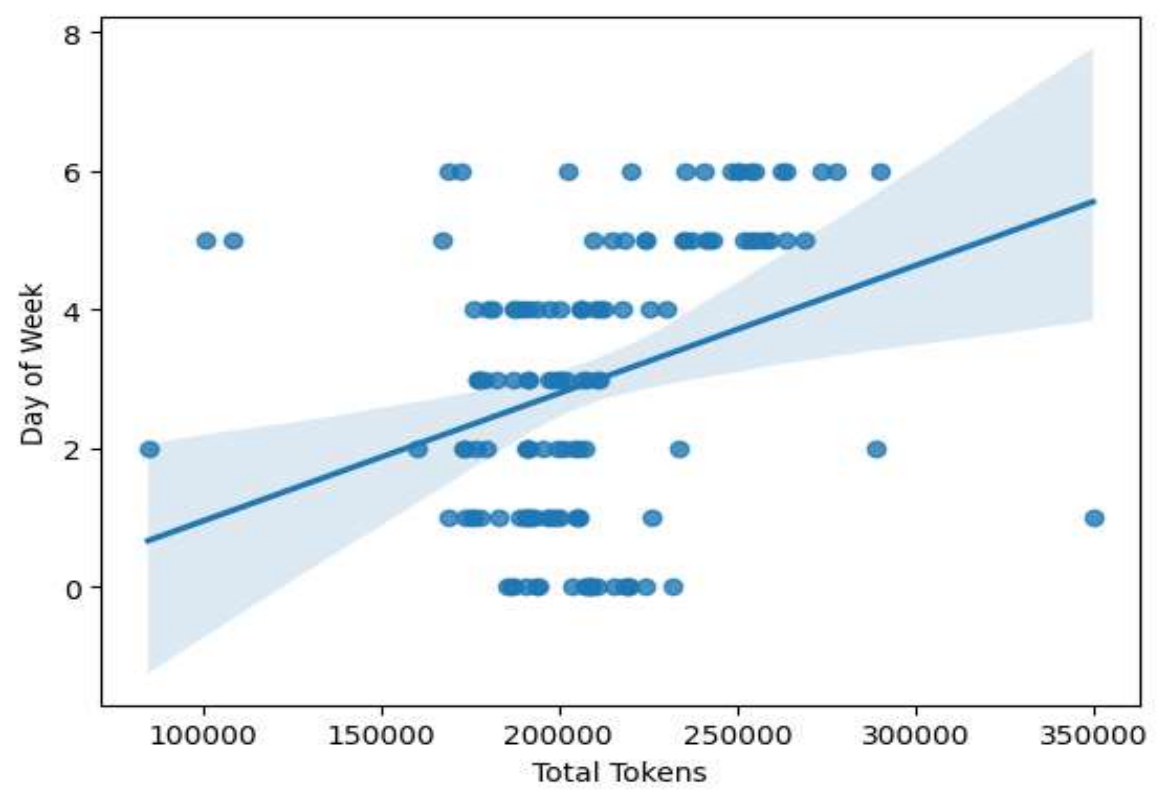
Every now and then, NammaMetro fails to report ridership numbers.

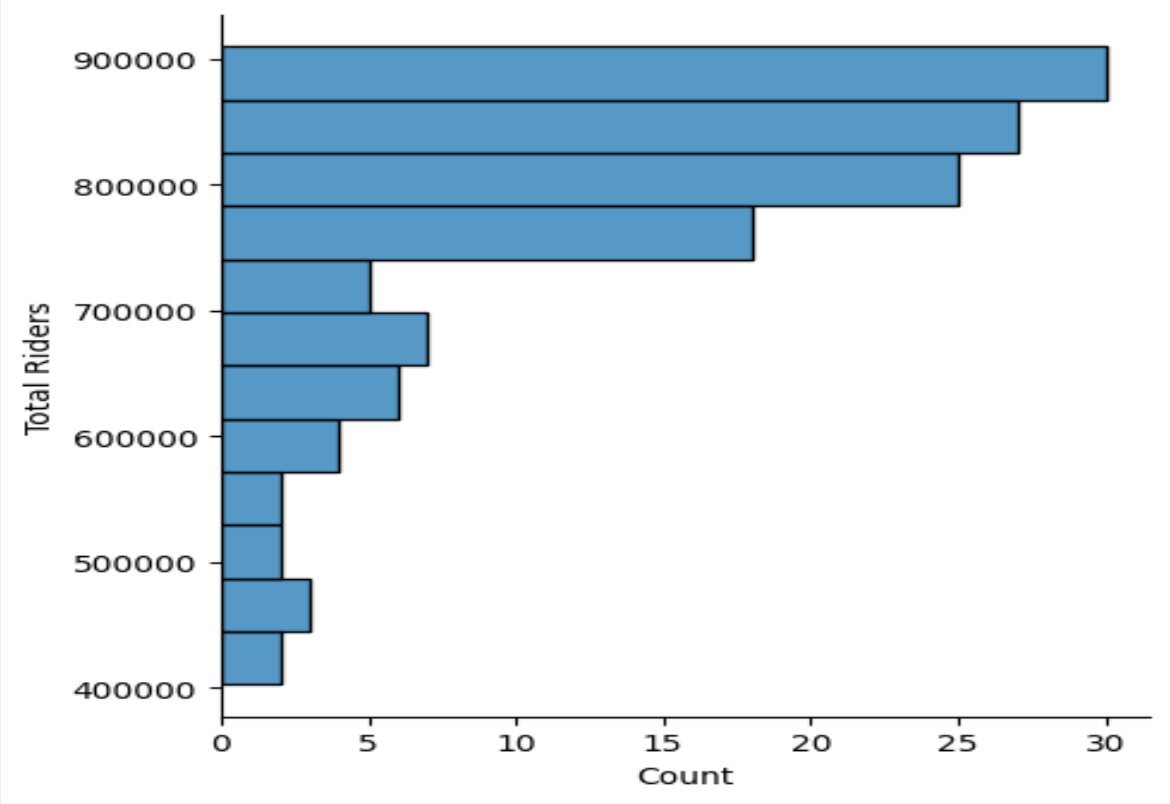
This usually happens when the individual tasked with manually updating (!!!) the data page goes on leave.

Let's get a better understanding of that missing data.

	Record Date	Day of Week	Total Riders
113	2025-02-16	Sunday	<NA>
120	2025-02-23	Sunday	<NA>
127	2025-03-02	Sunday	<NA>
137	2025-03-12	Wednesday	<NA>
138	2025-03-13	Thursday	<NA>
141	2025-03-16	Sunday	<NA>
142	2025-03-17	Monday	<NA>
150	2025-03-25	Tuesday	<NA>
152	2025-03-27	Thursday	<NA>
153	2025-03-28	Friday	<NA>

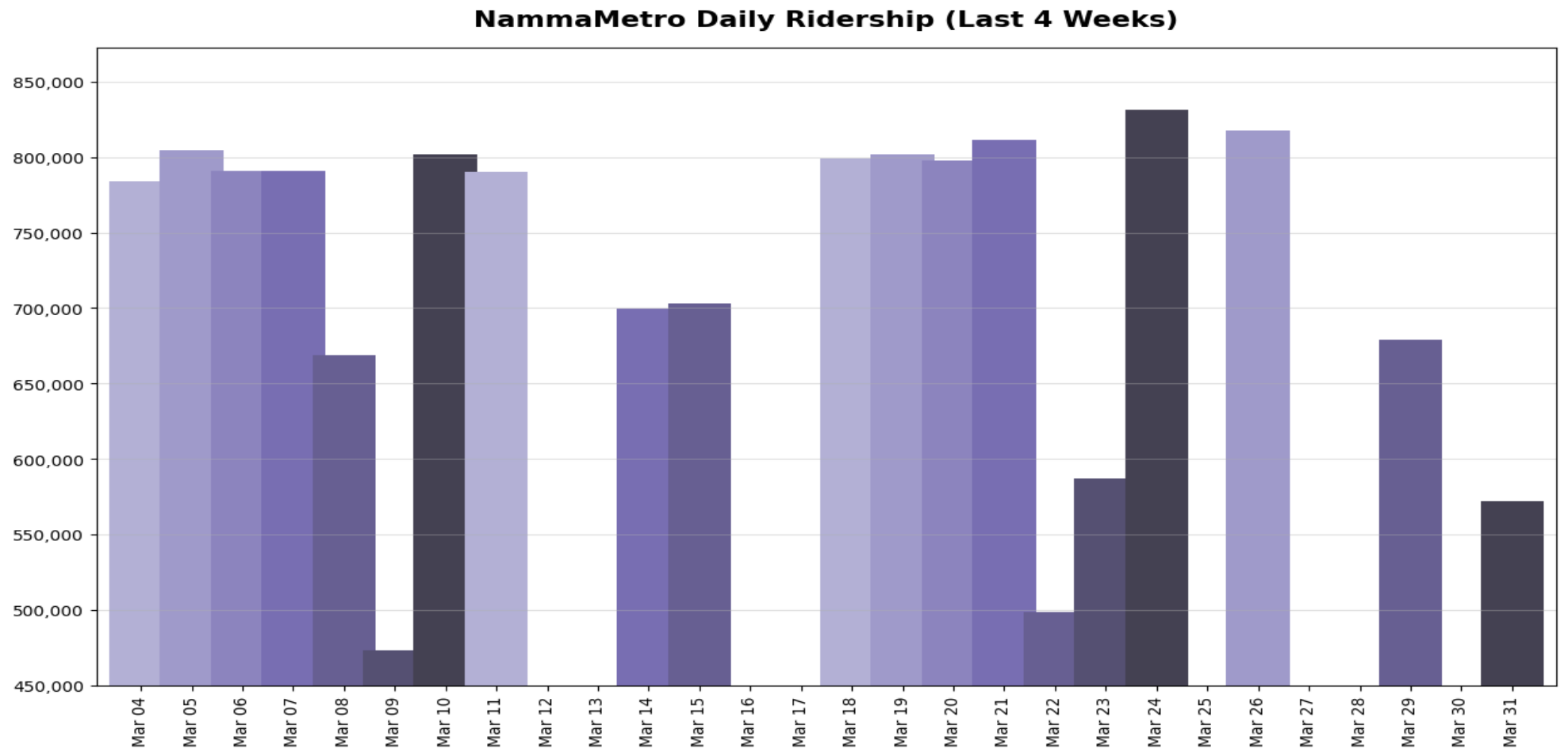
2. What is the overall trend in daily metro ridership over the selected period?





The trend shows a consistent rise in daily ridership with noticeable increases post infrastructure enhancements and fare structure changes.

3.What does the recent 4-week daily ridership bar chart reveal about commuter behavior?



Metro Is Getting Crowded

Top 10 Busiest Days (Commuter Traffic)

THE TIMES OF INDIA

City Bengaluru Mumbai Delhi Hyderabad Kolkata Chennai Agra Agartala Ahmedabad Ajmer ••• Today's ePaper

PHOTOS WEATHER KARNATAKA ELECTIONS

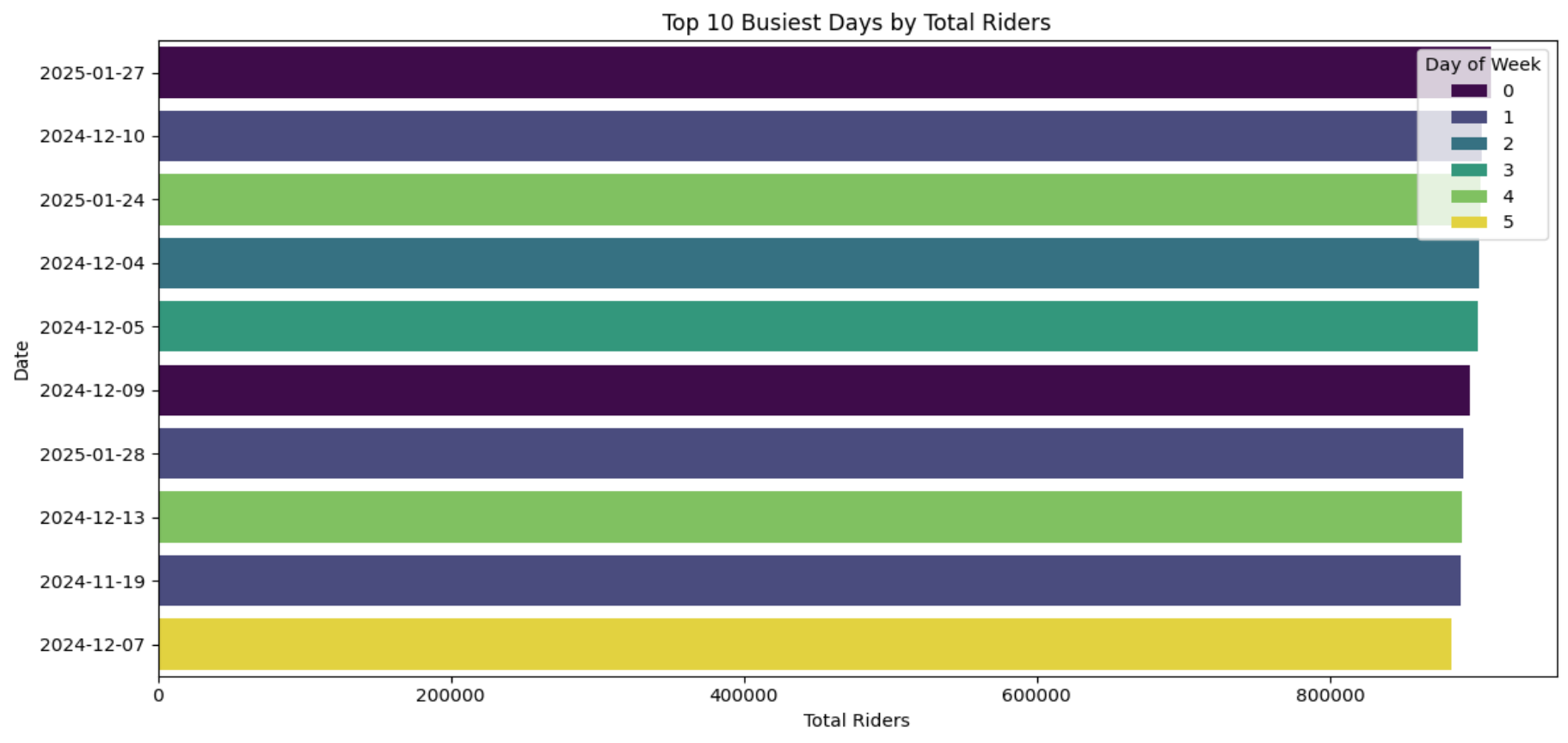
News · City News · Bengaluru News

THIS STORY IS FROM OCTOBER 13, 2023

With entire Purple Line operational, Bengaluru Metro ridership crosses 7-lakh mark

The ridership of Namma Metro in Bengaluru has increased to over 7 lakh after the launch of operations along the entire Purple Line. The average ridership before the opening of the missing links was 6.5 lakh. Commuters are demanding an increase in train frequency, especially during peak hours. BMRCL officials have stated that they are running adequate trains and have observed changes in footfall at different stations. The opening of the missing link has also helped ease traffic congestion in certain areas.

4.What patterns can be observed in the top 10 busiest days of metro ridership



👆 In December 2024, NammaMetro's single-day ridership crossed 900,000 for the first time! The ridership hit a record of 700,000 passengers in October 2023. It took just a little over a year to add 200,000 riders/day to the system.

One Week In NammaMetro

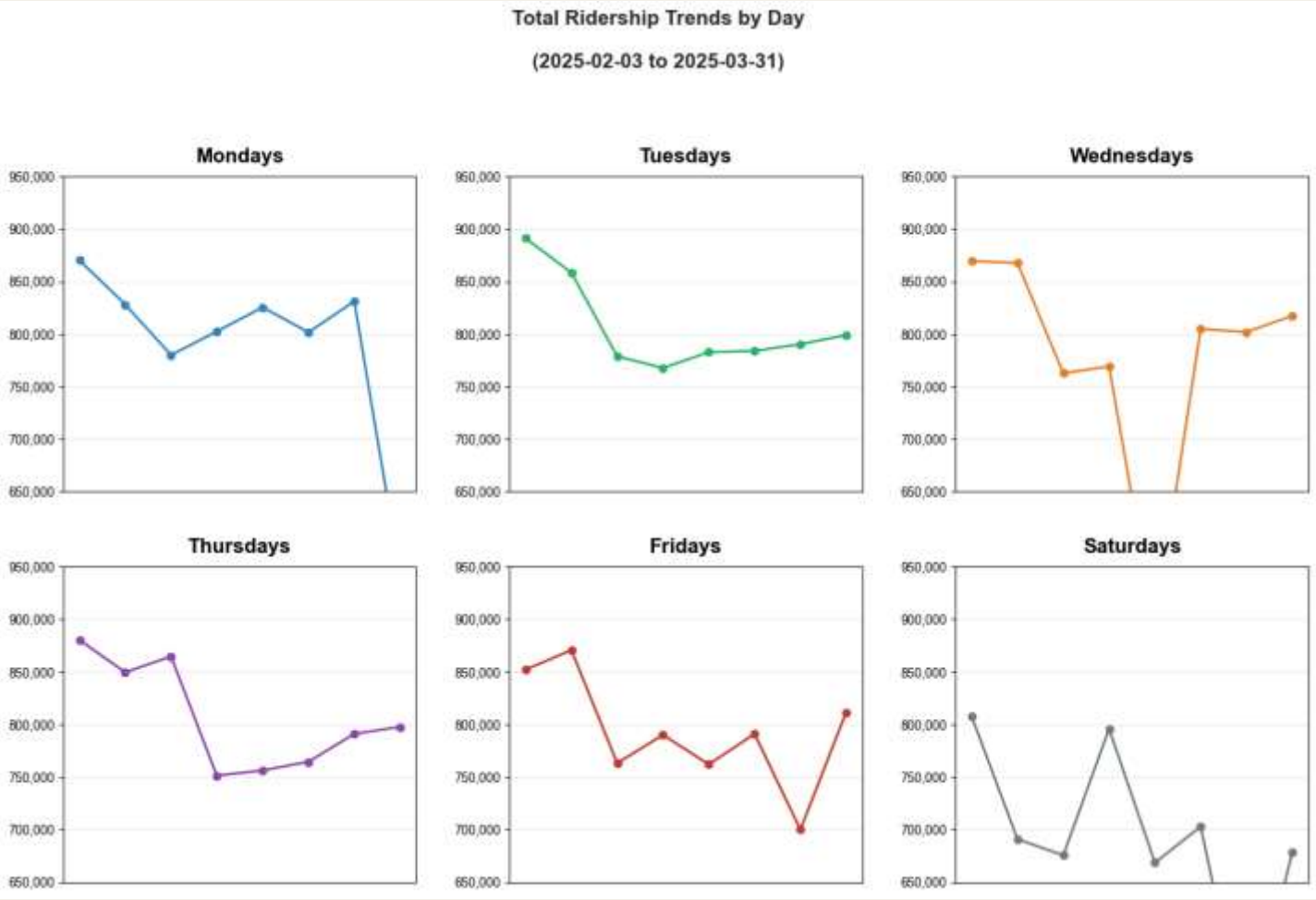
Total Ridership over the Last 7 Days

Total Ridership by Day of Week

Average Ridership by Day of Week

Ridership Patterns by Payment

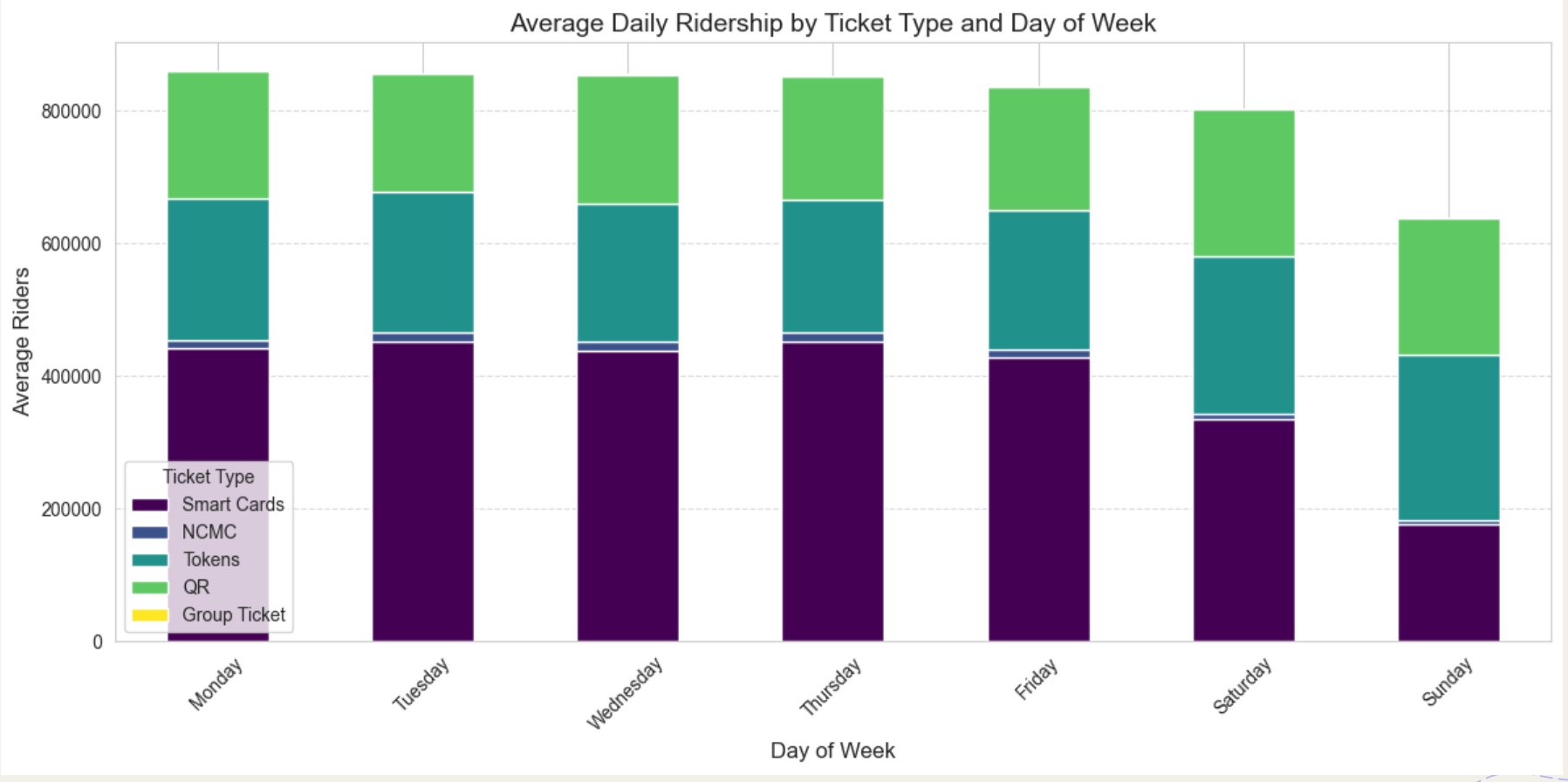
5.What do individual weekday ridership trends reveal about commuter behavior?



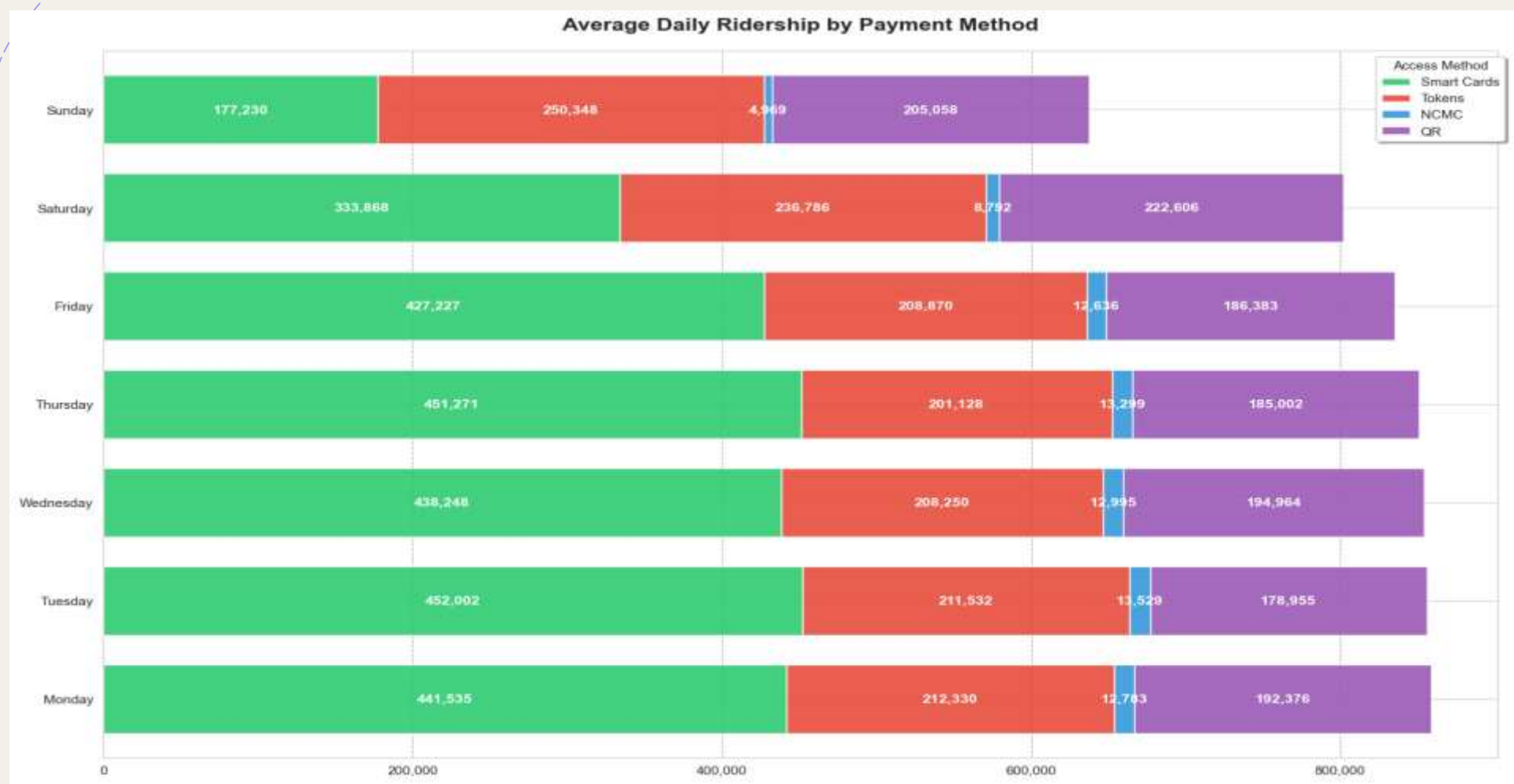
This is a series of line plots showing the trend in total ridership for each day of the week over the past several weeks. This makes it easy to spot patterns for specific days and answer several key questions concerning:

•consistency (*"Are there regular patterns in how ridership fluctuates?"*)
•trends (*"Are there seasonal or cyclical patterns in day-specific ridership?"*)
•anomalies (*"Which days experienced unusual spikes or dips?"*)

6.What insights can be drawn from the average daily ridership data grouped by day and ticket type?



7.How do different payment methods compare in daily usage?

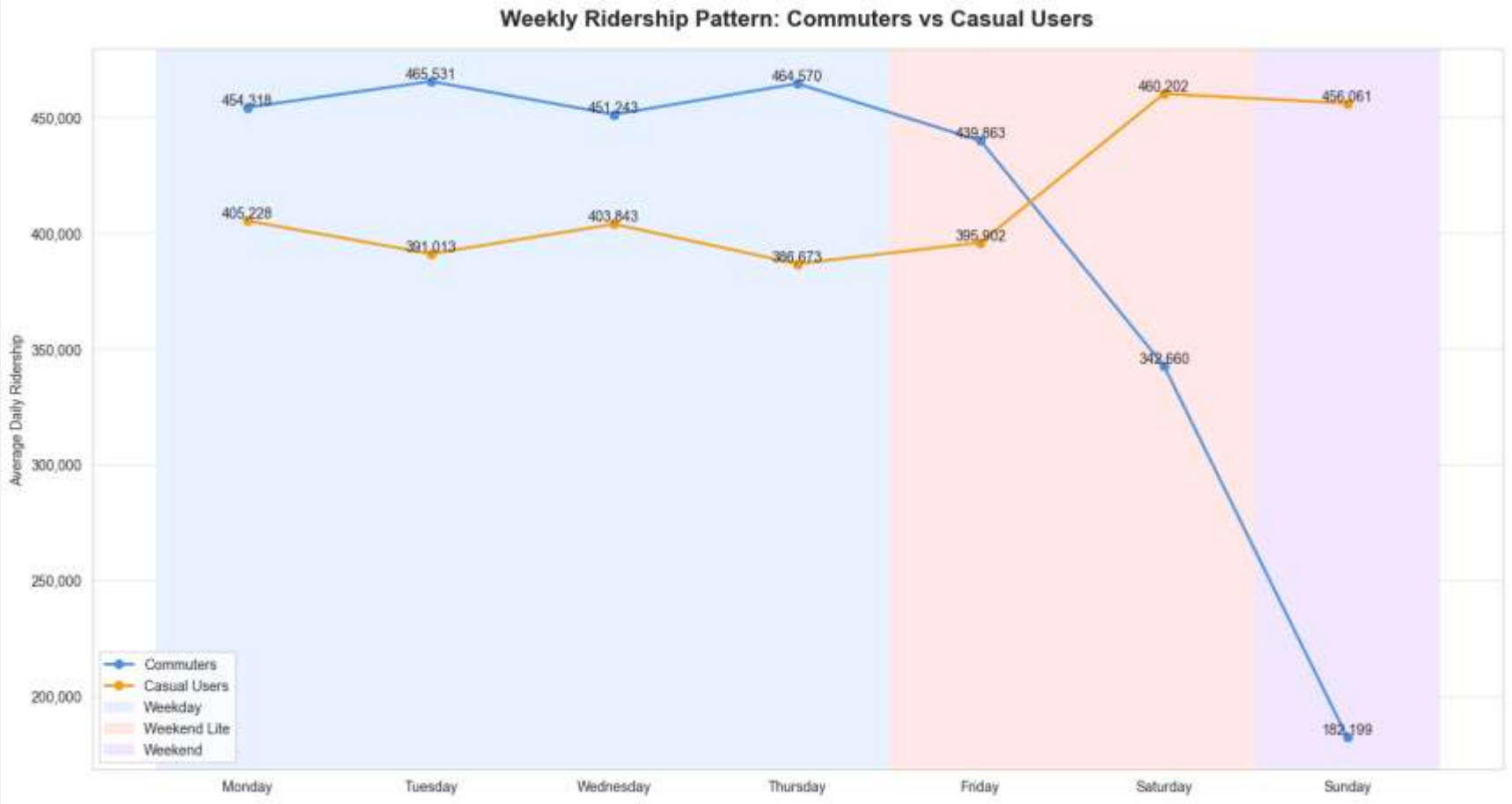


Smart Card and NCMC usage peaks during the week; Tokens and QR peaks on weekends.

Three Traffic Bands, Two Kinds Of Patron, One Metro

Ridership patterns: Commute vs Casual Rides by Traffic Band

8.What patterns emerge when comparing commuter and casual ridership across the week?



Two Kinds of Patrons

Ridership starts to dip on Fridays (perhaps due to liberal work-from-office policies in the IT industry or commuters preferring to use their personal vehicles.) For another segment of workers, Saturdays are as much a working day as the rest of the week.

On Sundays, Smart Cards+NCMC usage sees a **~60% drop** from average Weekday levels. Tokens+QR **increases by ~15%**.

This behaviour confirms the presence of two distinct types of users: **Commuters** and **Casual Users**.

Three Traffic Bands

Weekdays – *Just Another Manic Monday to Thursday*

During weekdays, defined as Monday to Thursday, commuter ridership dominates, showing a consistent pattern that reflects the city's work rhythm. These commuters rely on Smart Cards and NCMC to quickly move in and out of the NammaMetro system.

Weekend *Lite – Should I Stay or Should I Go?*

A remarkable crossover occurs in this band. Friday marks a clear shift where casual ridership begins to rise while commuter numbers decline.

Weekend – *A Day of Rest For Coders (But Not QR Coders)*

The starkest contrast appears on Sundays, where the two rider categories tell very different stories. Commuter numbers plummet to their lowest point while casual ridership reaches peak levels. This dramatic reversal reveals how the metro adapts to serve leisure travelers and weekend activities even as regular commuters largely stay home.

One Month On NammaMetro

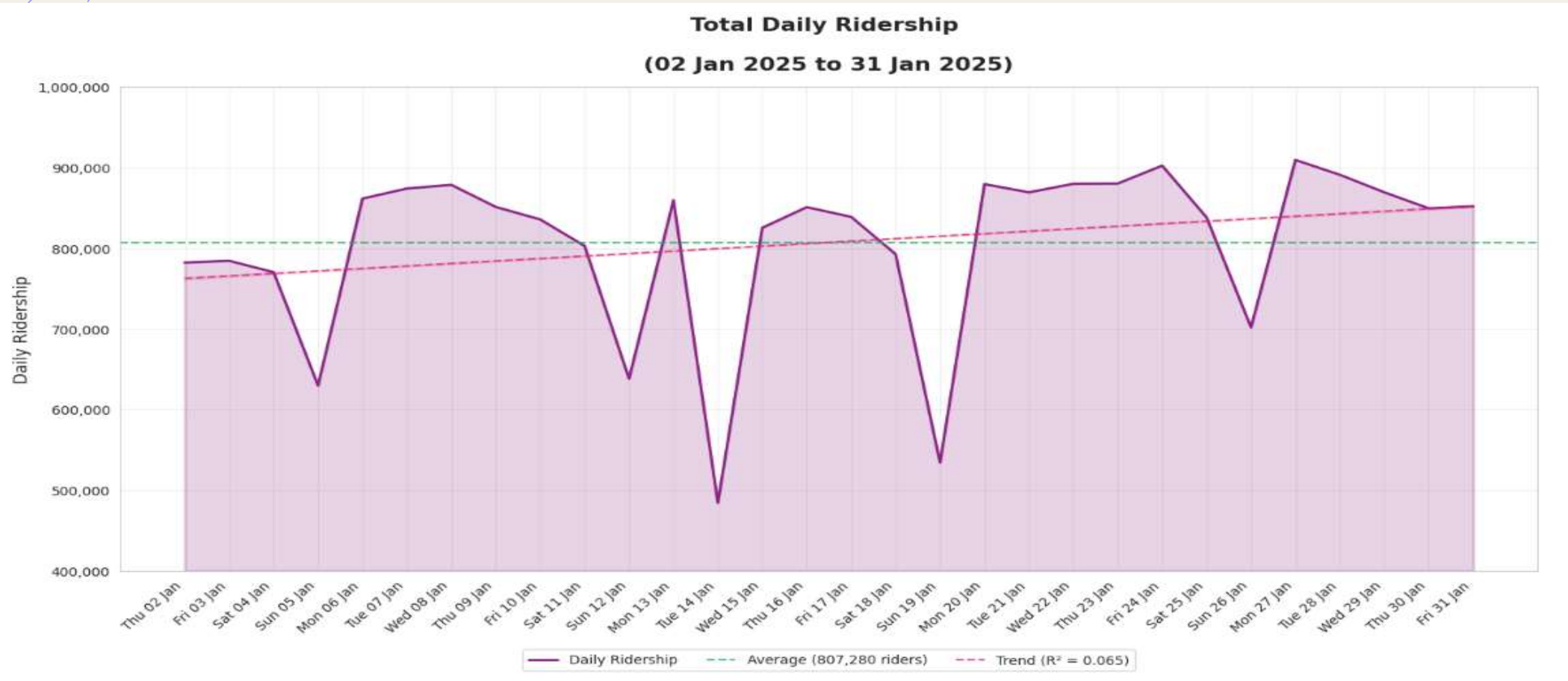
The Spread of Data

Ebb and Flow of Daily Passenger Traffic

A Wave Rides NammaMetro

9.What does the monthly ridership boxplot reveal about distribution, consistency, and anomalies?

10.What is the overall trend in daily metro ridership over the selected period?(EBB and Flow)



R² Interpretation Scale - Ranging from 0.0 to 1.0, R² is a measure of how well the regression line explains variability in ridership data.

11.How has the usage of One-Day, Three-Day, and Five-Day passes changed over time?

This visualisation shows a striking **inverse relationship between commuter and casual users** thus reinforcing a key finding about the dual nature of the metro system
The visualisation shows data gaps with dashed lines, maintaining data integrity while still allowing for trend analysis.

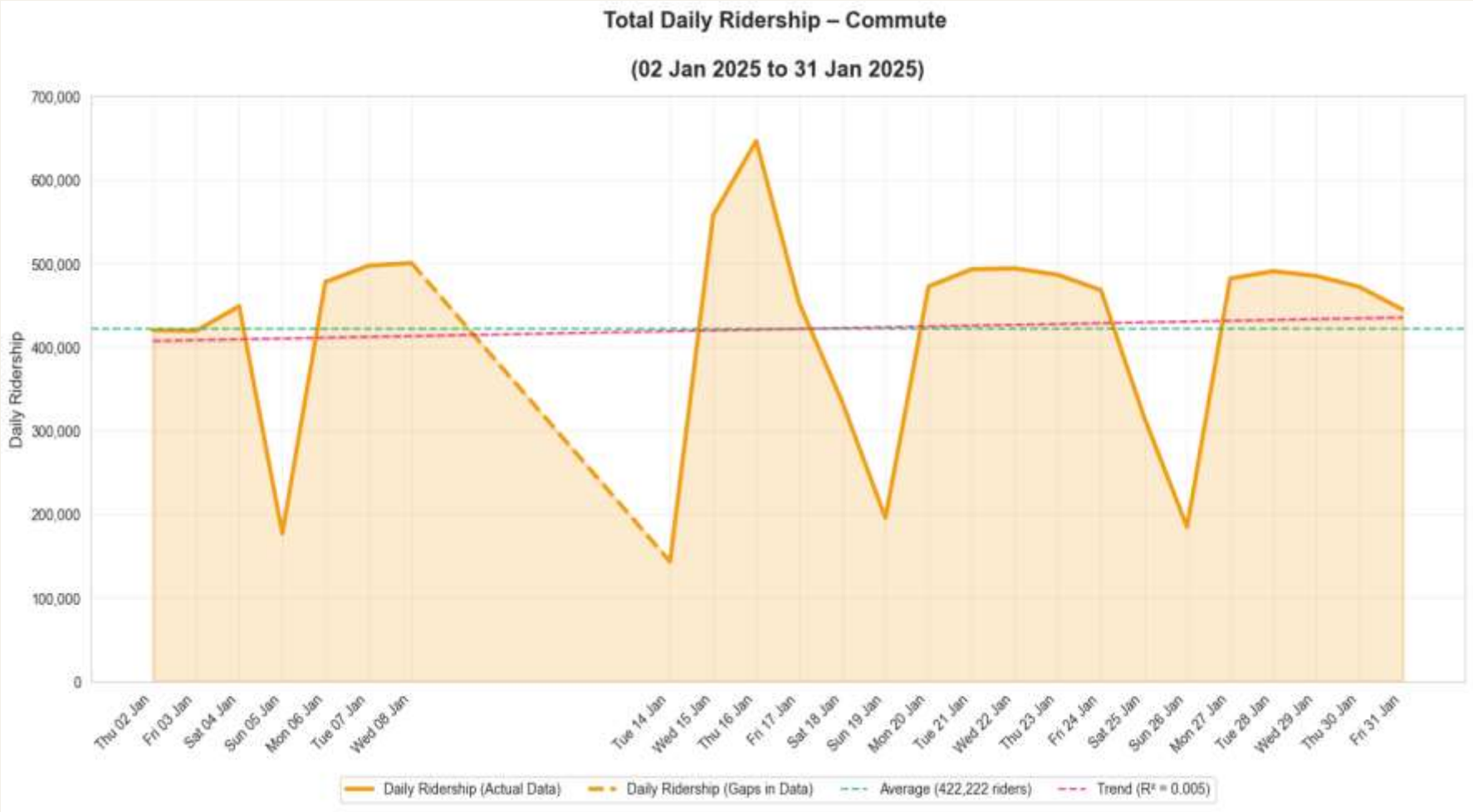
Long Weekend and Other Phenomena

Sankranti and Something Else

Ridership Towers – Users and Payments

12.What is the Total Ridership - Casual?

13.What is the Total Ridership - Commute?



Holidays and cultural events have dramatically different impacts on different user segments. The separate but complementary views of Commute and Casual ridership (seen above) expose patterns that would be completely hidden in the aggregate data.

1. **Inverse Relationship:** The separate ridership trends for Casual and Commute users confirm a pattern we've observed before: when Commuters drop, Casual riders rise. However, there are moments—like Sankranti—when this balance breaks down.

2. **Data Gap Handling:** Dashed lines indicate gaps in data collection. Showing these explicitly rather than interpolating them maintains data integrity.

3. **January 14: The Sankranti Effect**

- The festival caused a sharp drop in Commuters, as many left Bangalore to visit their hometowns.
- Unlike regular weekends, Casual ridership did not increase to compensate—suggesting an overall citywide slowdown.
- Compounding the disruption, Sankranti fell on a Tuesday, leading to an extended long weekend effect.

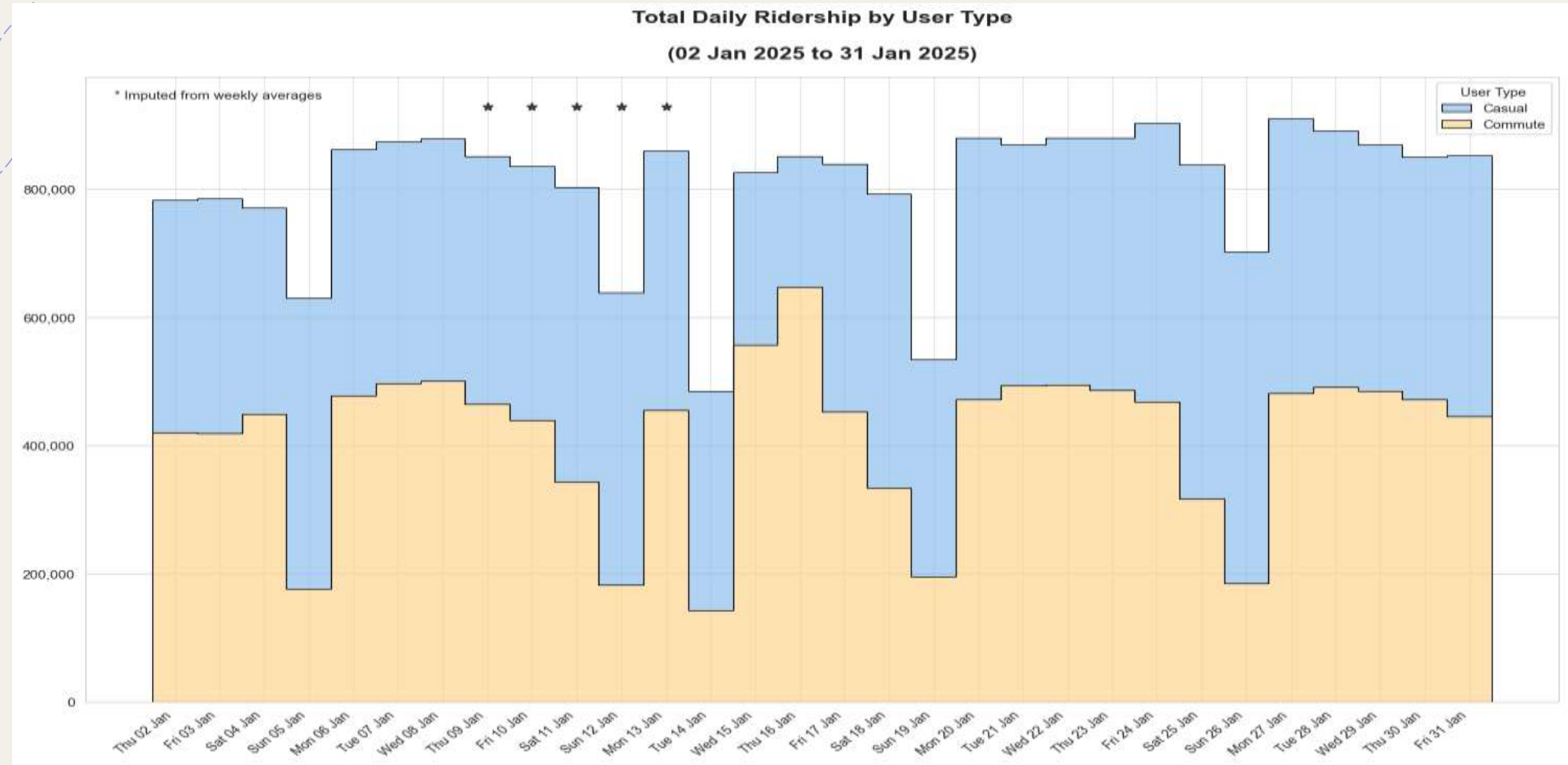
☞ Unfortunately, the data gap from Jan 13-17 prevents us from fully seeing this play out. It seems our data collector at NammaMetro also took a long weekend!

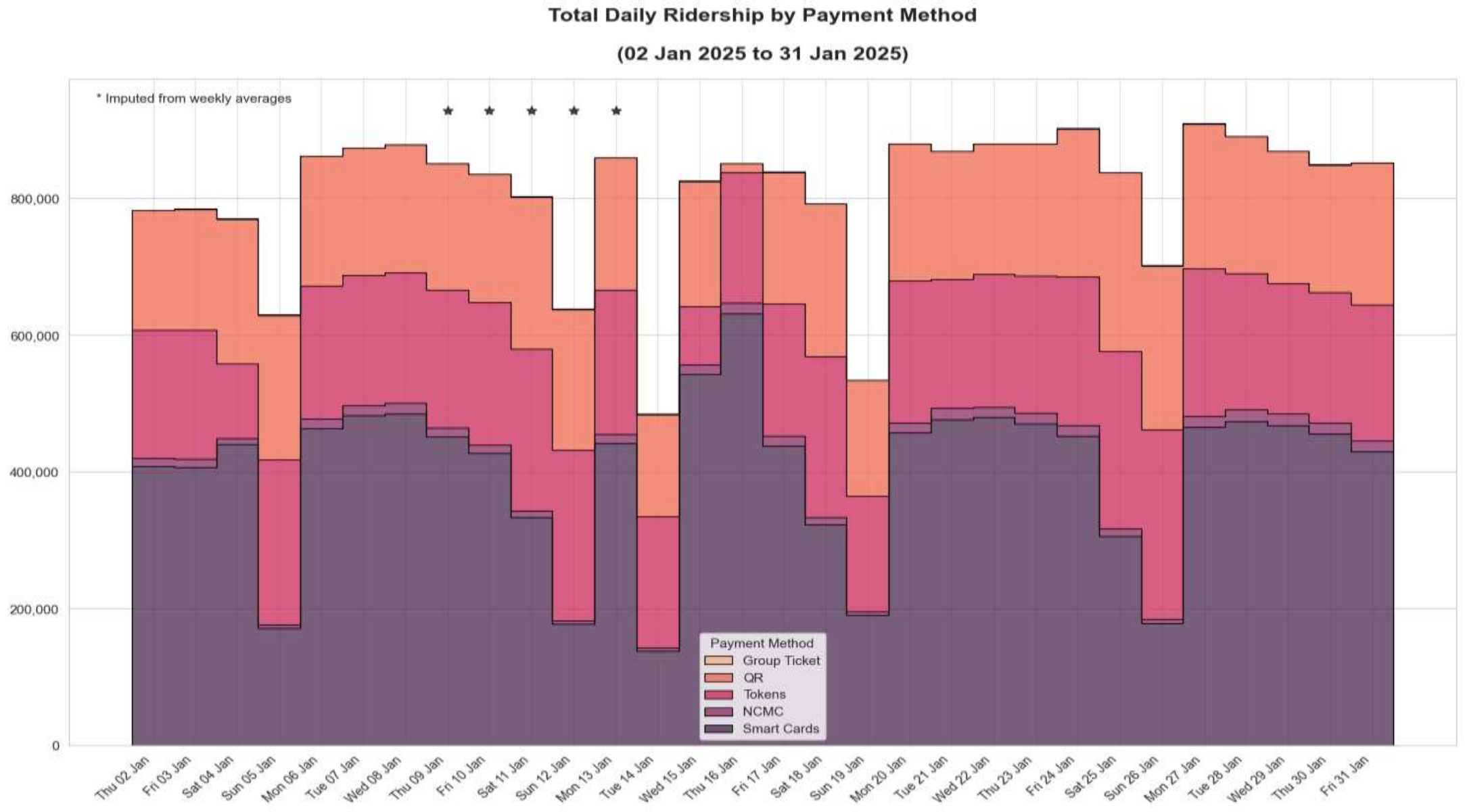
4. **Casual Ridership Growth vs. Commuter Stability:** $R^2 = 0.105$ indicates steady growth in Casual ridership, while the Commuter trend ($R^2 = 0.005$) remains largely flat. This hints at broader behavioral shifts—possibly an **increase in tourism or recreational metro usage**.

5. **Weekend Pattern Clarity:** The weekend dips in Commute ridership are more pronounced than the corresponding peaks in Casual ridership, explaining why the total ridership still shows net dips on weekends despite the casual user increase.

 What remains now is to explain the less obvious bump in **Commuter traffic on Jan 15-16** and **Casual riders on Jan 25**.

14.How do commute users differ from casual users in terms of ridership volume?




Total Daily Ridership by Payment Method
(02 Jan 2025 to 31 Jan 2025)
* Imputed from weekly averages
800,000
600,000
400,000
200,000
0
Payment Method
Group Ticket
QR
Tokens
NCMC
Smart Cards
Thu 02 Jan
Fri 03 Jan
Sat 04 Jan
Sun 05 Jan
Mon 06 Jan
Tue 07 Jan
Wed 08 Jan
Thu 09 Jan
Fri 10 Jan
Sat 11 Jan
Sun 12 Jan
Mon 13 Jan
Tue 14 Jan
Wed 15 Jan
Thu 16 Jan
Fri 17 Jan
Sat 18 Jan
Sun 19 Jan
Mon 20 Jan
Tue 21 Jan
Wed 22 Jan
Thu 23 Jan
Fri 24 Jan
Sat 25 Jan
Sun 26 Jan
Mon 27 Jan
Tue 28 Jan
Wed 29 Jan
Thu 30 Jan
Fri 31 Jan

1. These plots visualise the **scale of metro operations** (serving hundreds of thousands daily) and the **relative popularity of different payment options** while also highlighting the **strong weekly seasonality in urban transit usage**.

2. The asterisks at the top of the chart (indicating imputed data) also highlight how data collection gaps can impact analysis - particularly unfortunate during the Sankranti period when we might have seen interesting cultural patterns.

3. These visualisations effectively demonstrate how the metro serves dual purposes as both essential infrastructure for commuters and as a convenience for casual/leisure travelers, with the balance shifting dramatically throughout the week.

January 15-16 Anomaly:

Hidden beneath a rather normal-looking Thursday ridership, we see the shrinkage in QR+Tokens ("Casual") is compensated by a boost in Smart Card+NCMC ("Commute").

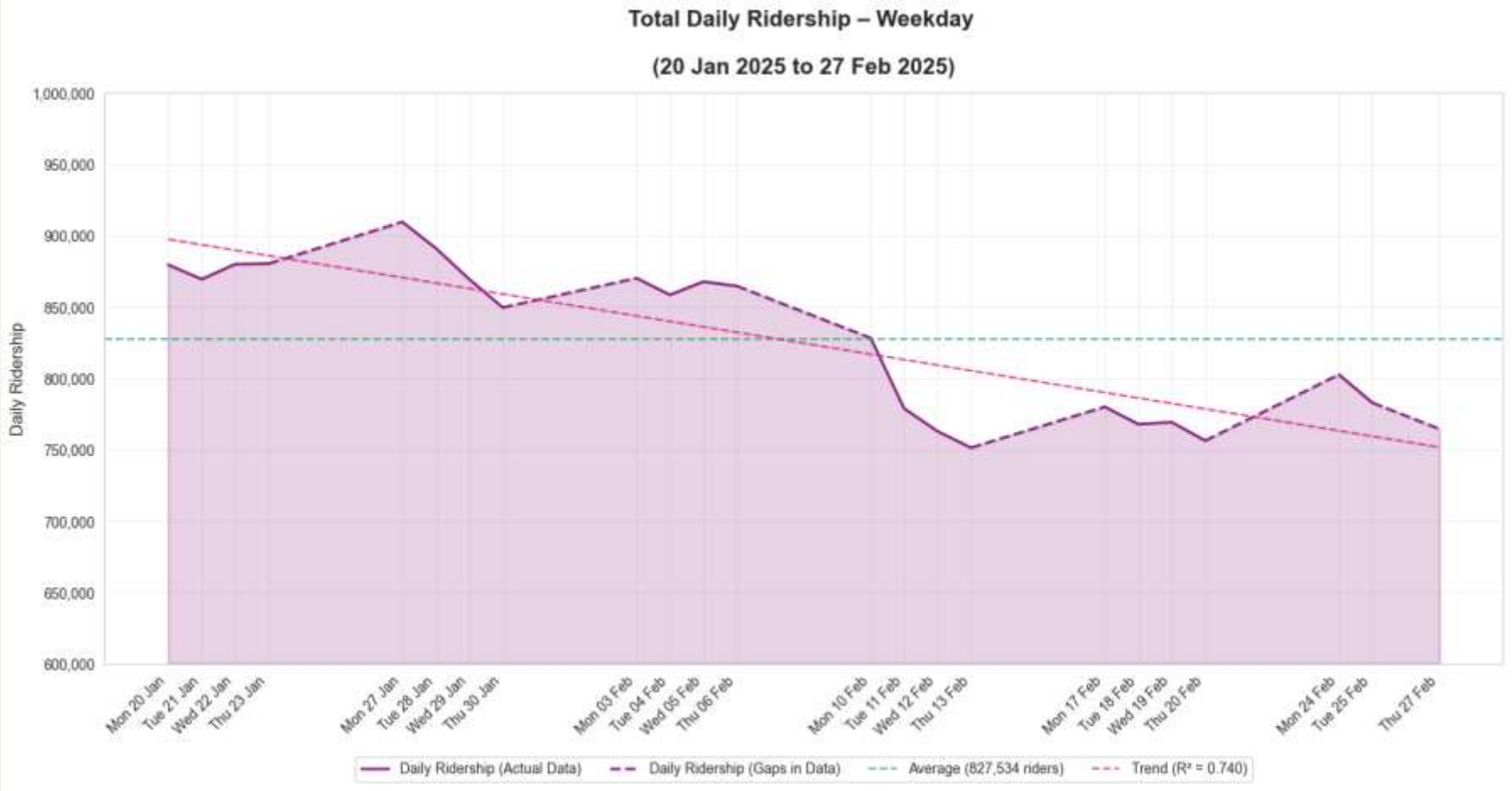
Upon closer observation, we see that QR payments were close to zero, indicating possible system disruption. Consequently, it appears that disappointed users have opted for Smart Cards to get on with their day.

Fare Hike of February 2025

Did users abandon NammaMetro in the days following the fare hike?

The Bigger Picture: Migration vs. Abandonment

15.How did the Fare hike affect the Total Ridership?



The data presents a **clear downward trend** in daily weekday ridership, with a **trendline (R^2 = 0.740)** indicating a strong fit.

After the Hike

✓ **Ridership decline is evident**

- The trendline suggests a sustained drop rather than a random fluctuation.

✓ **Sharp immediate fall**

- In the week following the fare hike, ridership plunged significantly, suggesting a strong commuter reaction.

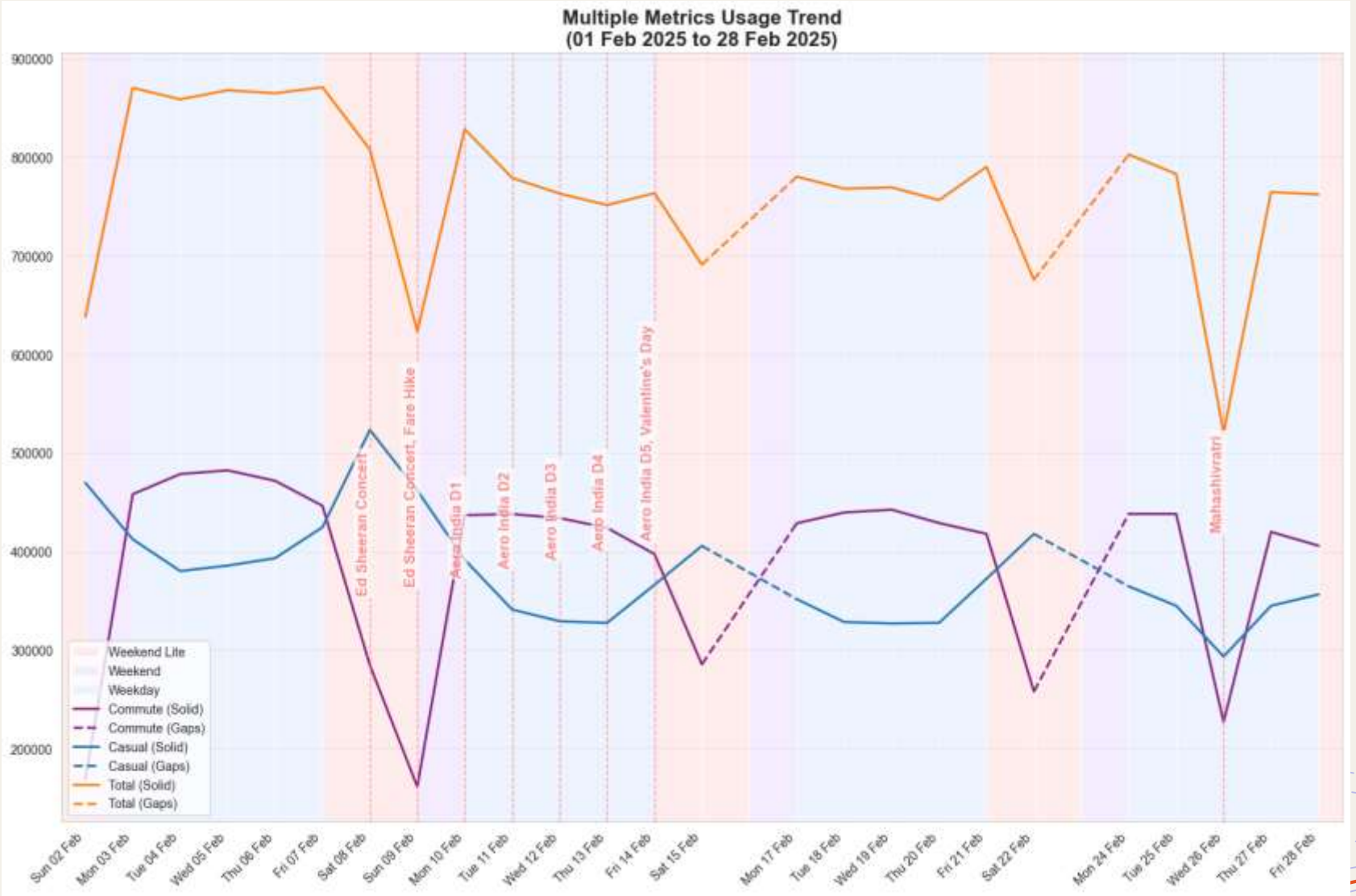
✓ **Quantifiable drop**

- In the three weeks before Feb 9, the **average weekday ridership** was **874,000 riders**.
- In the three weeks after, it fell to **777,000 riders**, an **11.1% decrease**.

✓ **No immediate recovery**

Ridership does not return to previous levels, suggesting a **short-term sustained impact**.

16.Did certain events or Holidays have an impact?



17.Are weekends before or after events showing unusual ridership behavior?

⚠ **Payment method trends reveal a shift**

- **Smart Card** users remain the majority, but the decline in token-based payments suggests infrequent riders reduced trips.
- A **drop in QR and NCMC**-based transactions indicates a potential pullback from occasional users or tourists.

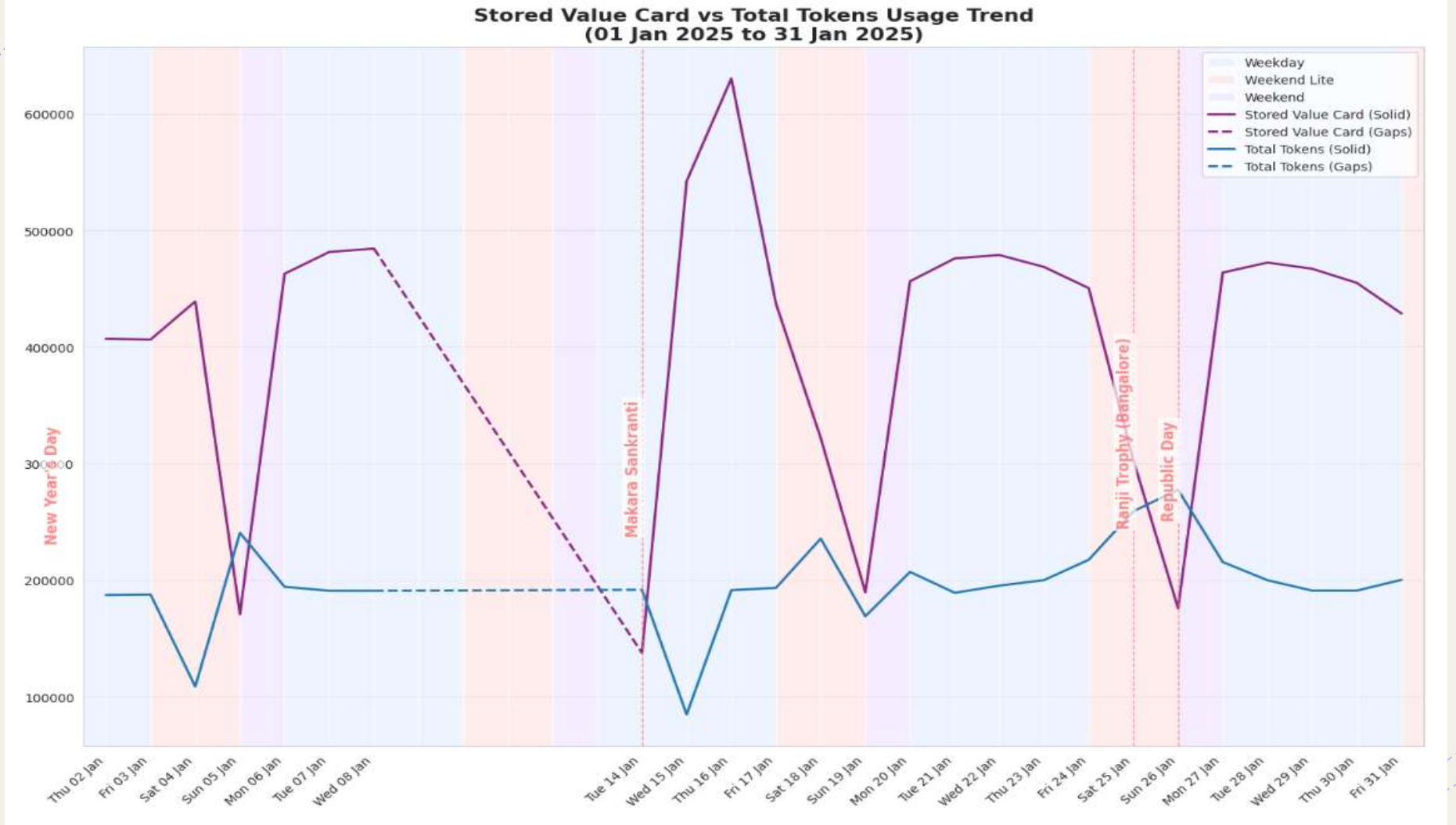
⚠ **Fare structure changes may have influenced behavior**

- Along with the fare hike, BMRCL introduced new fare rules:
    - **Smart Card & NCMC** users get a 5% discount over token fares.
    - During off-peak hours (before 8am, 12pm-4pm, after 9pm), a 10% discount is applied.
    - **QR payments** don't get any discounts, potentially discouraging casual users.
- This may explain why QR-based transactions dropped while Smart Card usage remained stable.

⚠ **A closer look at trends**

- The ridership trends are not solely explained by the fare hike—February saw major events impacting travel patterns.
- **Aero India** (Feb 10-14, 2025) likely pulled potential metro users to an area outside the network, further depressing ridership.
- The **Ed Sheeran concerts** (Feb 9-10) fueled a final pre-hike surge**, followed by a steep drop.
- **Mahashivaratri** (Feb 26) introduced further fluctuations, particularly in casual travel.

18.Is there a visible trend in QR usage across multiple sources (Paytm, WhatsApp, etc.)?



Jan 15: Notice the sudden **drop in Tokens** and sharp **rise in Smart Cards**.

Jan 16: A **collapse in QR payments** is accompanied by an **astronomical surge in Smart Cards**.

Conclusion

The Bangalore Metro Ridership project presents a comprehensive view of how the city's metro system is being utilized, evolving, and influenced by external factors. Through careful preprocessing and visualization of ridership data, several key insights have emerged:

1. Ridership Growth is Consistent and Healthy
Over time, daily ridership has shown a steady upward trend, supported by infrastructure expansion, improved connectivity, and the increasing preference for public transport among city dwellers.

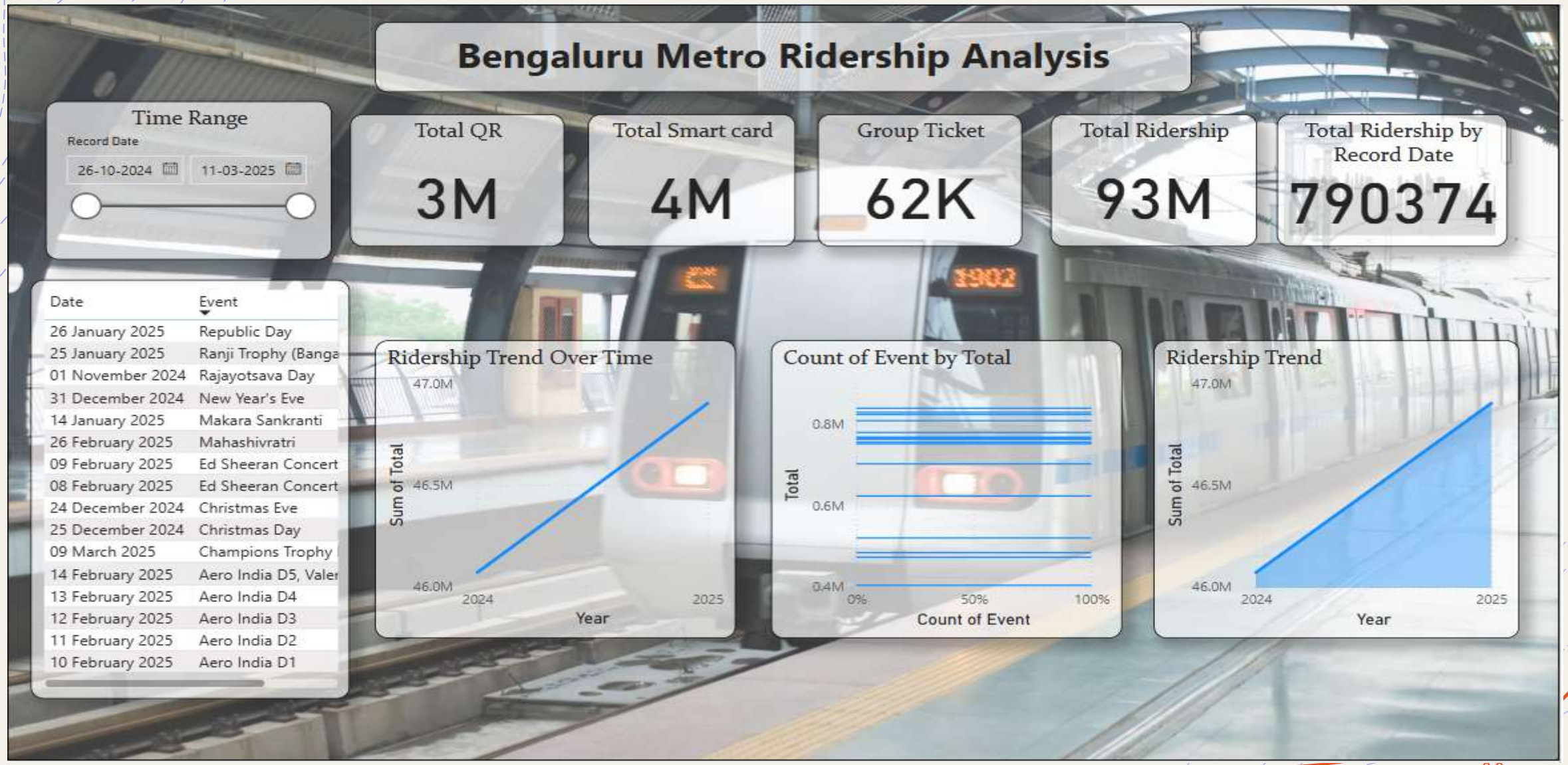
2. Commuters Dominate Weekdays, Casuals Rise on Weekends
The breakdown by user type indicates that regular commuters are the primary drivers of weekday traffic, while casual usage sees spikes during weekends, holidays, and events. This behavior suggests a stable core ridership with occasional inflows based on situational needs.

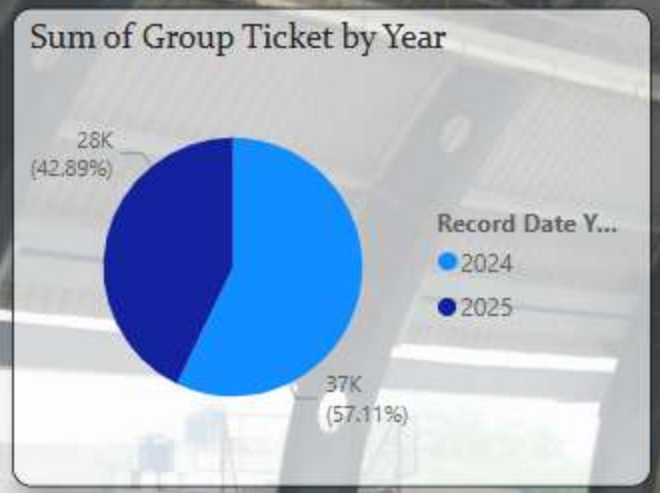
3. Events and Fare Hikes Have Tangible Effects
Policy changes, such as the February 9 fare hike, had short-term impacts – particularly reducing casual usage. Conversely, system expansions and long weekends typically boosted ridership, as seen in outliers and post-event trends.

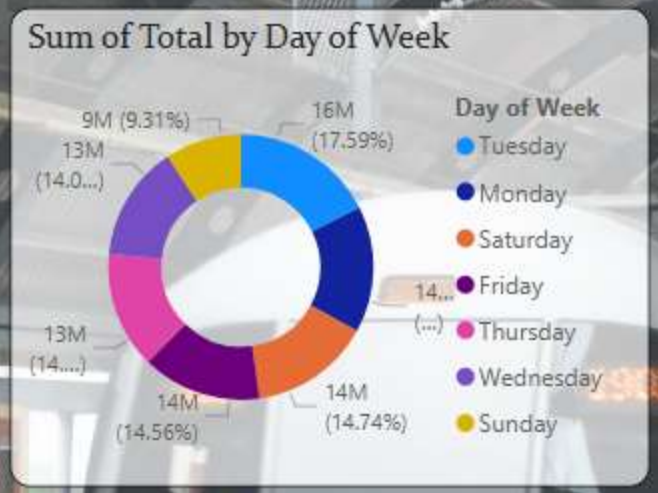
4. Payment Preferences Are Shifting
Smart Cards remain the most preferred payment method, especially among frequent users. QR-based ticketing is rising slowly but steadily, while traditional tokens and Group Tickets are on the decline, signaling a digital shift.

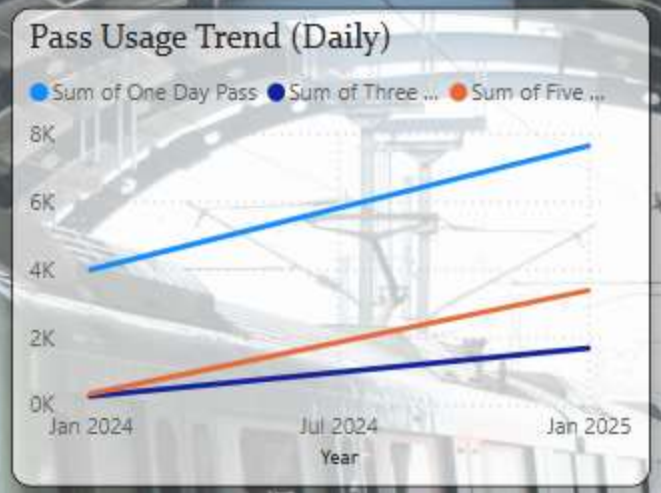
This analysis not only decodes how people move across Bangalore but also acts as a foundational framework for forecasting, policy testing, and operational optimization. With future integration of real-time data and geospatial mapping, it can evolve into a dynamic metro intelligence system.

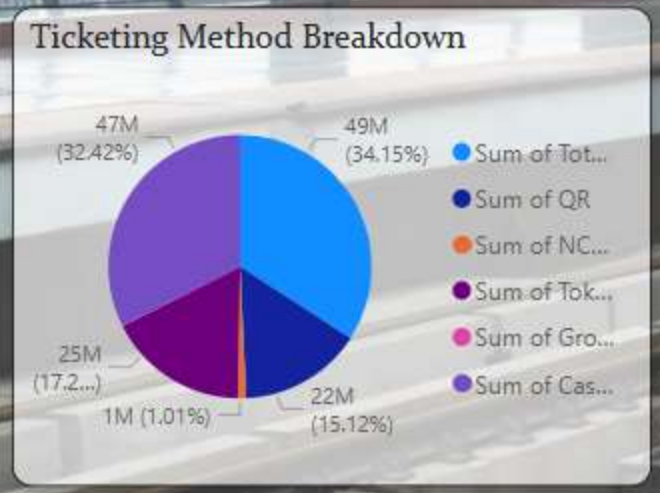
POWERBI

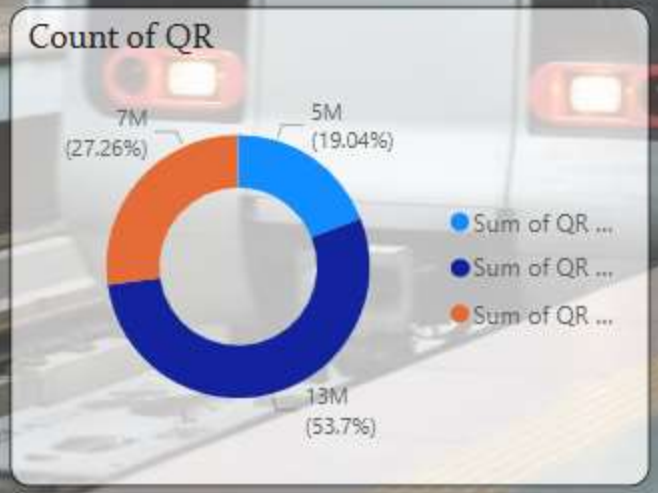

Bengaluru Metro Ridership Analysis
Time Range
Record Date
26-10-2024
11-03-2025
Total QR
3M
Total Smart card
4M
Group Ticket
62K
Total Ridership
93M
Total Ridership by Record Date
790374
Date
Event
26 January 2025
Republic Day
25 January 2025
Ranji Trophy (Banga
01 November 2024
Rajayotsava Day
31 December 2024
New Year's Eve
14 January 2025
Makara Sankranti
26 February 2025
Mahashivratri
09 February 2025
Ed Sheeran Concert
08 February 2025
Ed Sheeran Concert
24 December 2024
Christmas Eve
25 December 2024
Christmas Day
09 March 2025
Champions Trophy
14 February 2025
Aero India D5, Valer
13 February 2025
Aero India D4
12 February 2025
Aero India D3
11 February 2025
Aero India D2
10 February 2025
Aero India D1
Ridership Trend Over Time
47.0M
46.5M
46.0M
Sum of Total
2024
2025
Year
Count of Event by Total
0.8M
0.6M
0.4M
Total
0%
50%
100%
Count of Event
Ridership Trend
47.0M
46.5M
46.0M
Sum of Total
2024
2025
Year

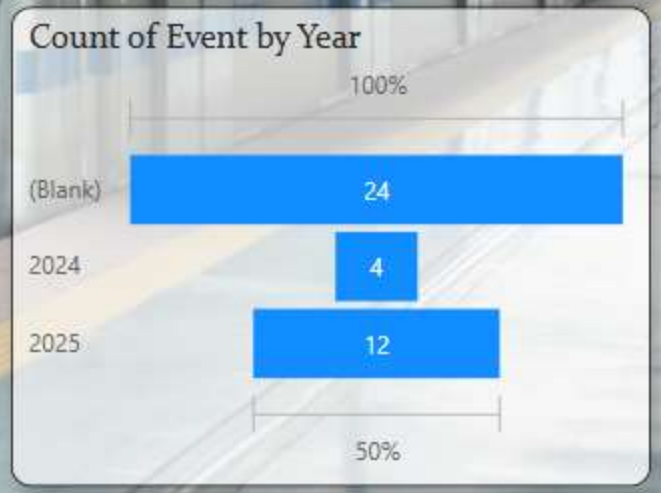

Sum of Group Ticket by Year
28K (42.89%)
37K (57.11%)
Record Date Y...
2024
2025

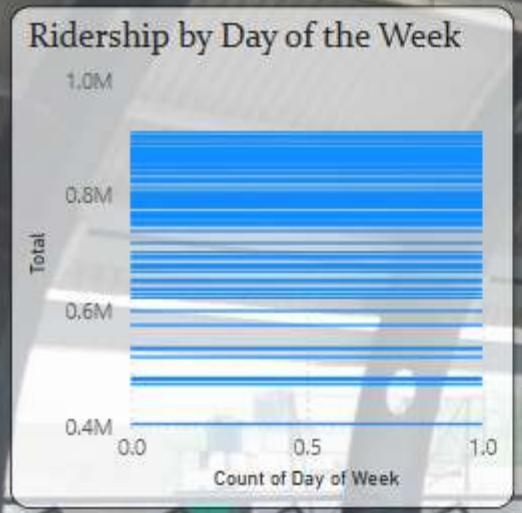

Sum of Total by Day of Week
9M (9.31%)
16M (17.59%)
13M (14.0...)
14... (...)
13M (14....)
14M (14.56%)
14M (14.74%)
Day of Week
Tuesday
Monday
Saturday
Friday
Thursday
Wednesday
Sunday

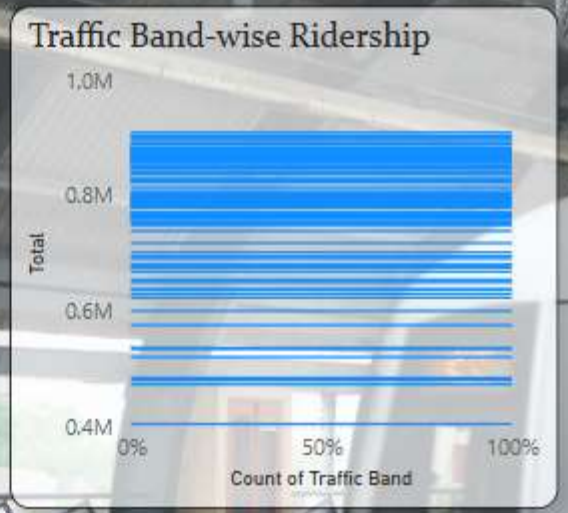

Pass Usage Trend (Daily)
Sum of One Day Pass
Sum of Three ...
Sum of Five ...
8K
6K
4K
2K
0K
Jan 2024
Jul 2024
Jan 2025
Year

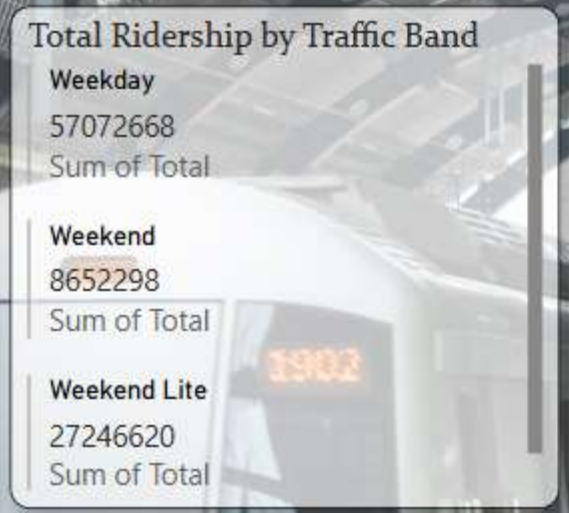

Ticketing Method Breakdown
47M (32.42%)
49M (34.15%)
25M (17.2...)
1M (1.01%)
22M (15.12%)
Sum of Tot...
Sum of QR
Sum of NC...
Sum of Tok...
Sum of Gro...
Sum of Cas...

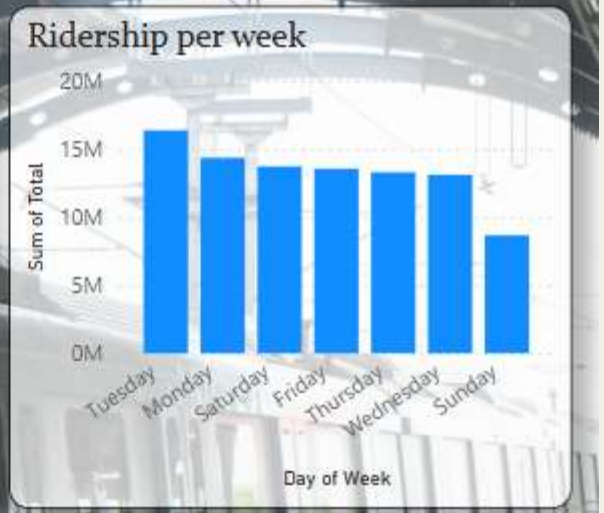

Count of QR
7M (27.26%)
5M (19.04%)
13M (53.7%)
Sum of QR ...
Sum of QR ...
Sum of QR ...

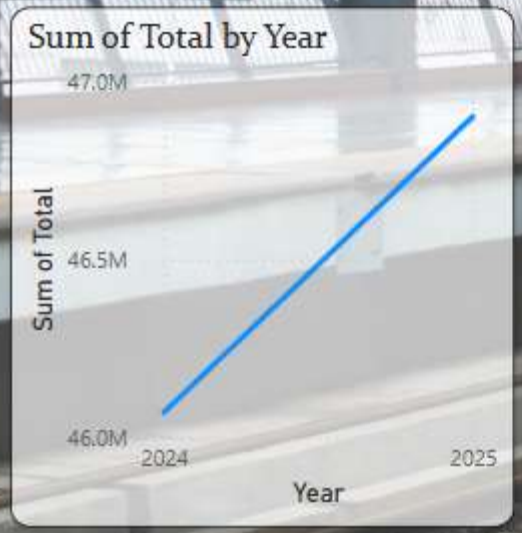

Count of Event by Year
100%
(Blank)
24
2024
4
2025
12
50%


Ridership by Day of the Week
1.0M
0.8M
0.6M
0.4M
Total
0.0
0.5
1.0
Count of Day of Week

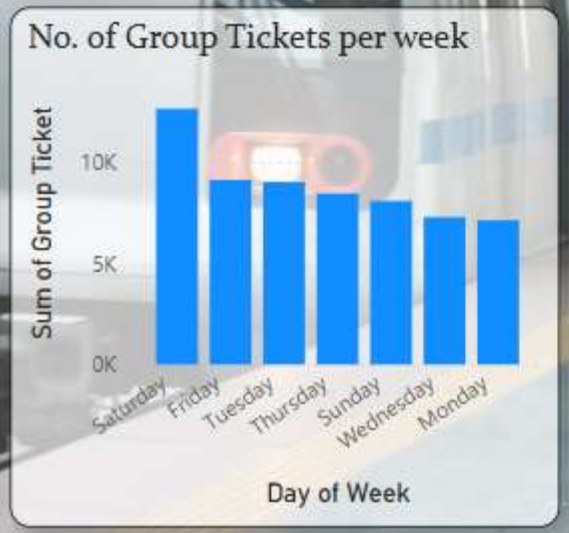

Traffic Band-wise Ridership
1.0M
0.8M
0.6M
0.4M
Total
0%
50%
100%
Count of Traffic Band

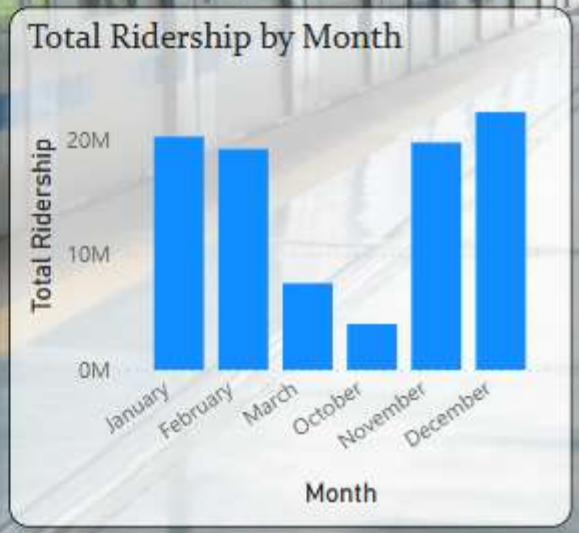

Total Ridership by Traffic Band
Weekday
57072668
Sum of Total
Weekend
8652298
Sum of Total
Weekend Lite
27246620
Sum of Total


Ridership per week
20M
15M
10M
5M
0M
Sum of Total
Tuesday
Monday
Saturday
Friday
Thursday
Wednesday
Sunday
Day of Week


Sum of Total by Year
47.0M
46.5M
46.0M
Sum of Total
2024
2025
Year

Ridership Trend per Day
4.1M
4.1M
4M
3M
2M
2.7M
2.5M
1.8M
1.5M
Sum of Total
0
20
40
Day


No. of Group Tickets per week
10K
5K
0K
Sum of Group Ticket
Saturday
Friday
Tuesday
Thursday
Sunday
Wednesday
Monday
Day of Week


Total Ridership by Month
20M
10M
0M
Total Ridership
January
February
March
October
November
December
Month

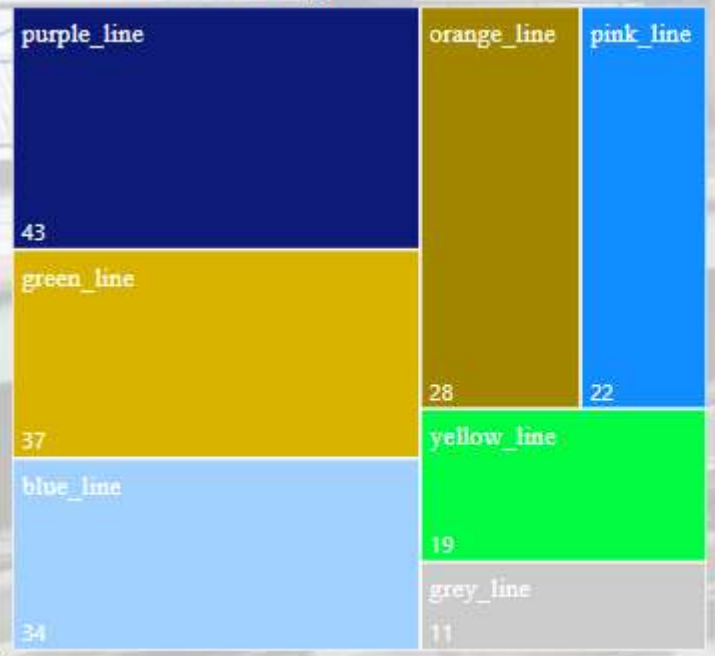
METRO STATION ANALYSIS

Station Name

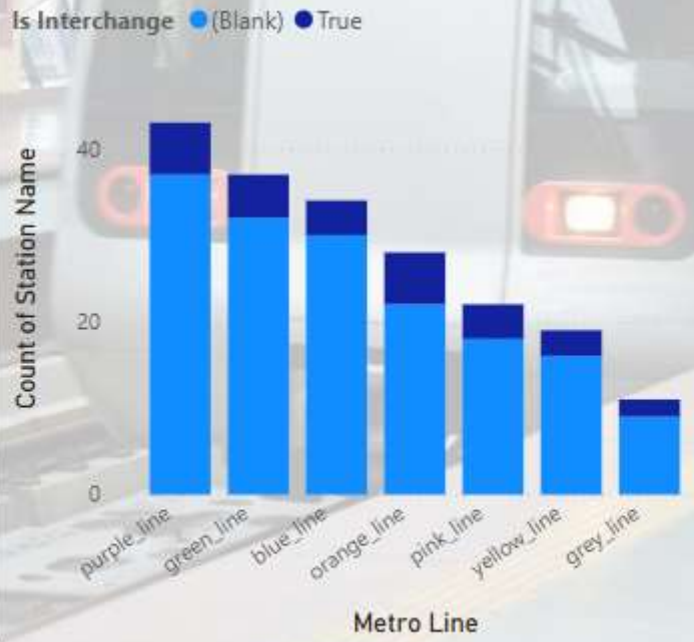
All

Station Name	Metro Line	Is Interchange
Yeshwanthpur	green_line	
Yelahanka (TBC)	blue_line	
Yelachenahalli	green_line	
Whitefield (Kadugodi)	purple_line	
Vinayaka Nagara (TBC)	grey_line	

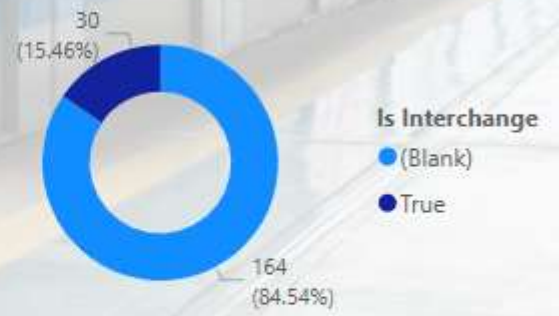
Number of Stations per Line



Interchange vs. Regular Stations by Line



Interchange vs Non-Interchange Split



Thank You